

Chemical Equipment Analysis Report

Dataset: sample40.csv
Upload Date: 2026-02-03 15:19
Total Equipment Count: 40

Summary Statistics

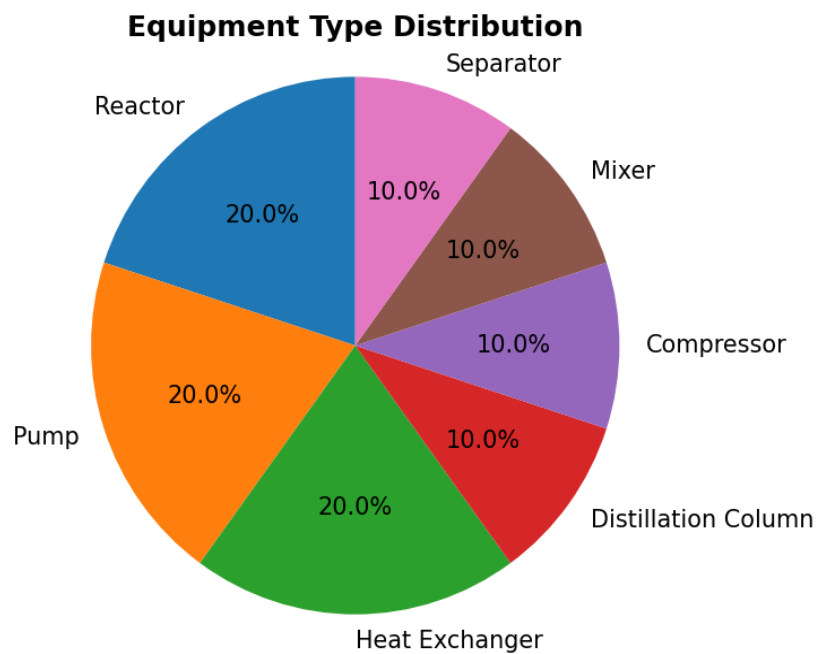
Metric	Value	Unit
Average Flowrate	188.06	L/min
Average Pressure	58.33	psi
Average Temperature	104.72	°C

Equipment Type Distribution

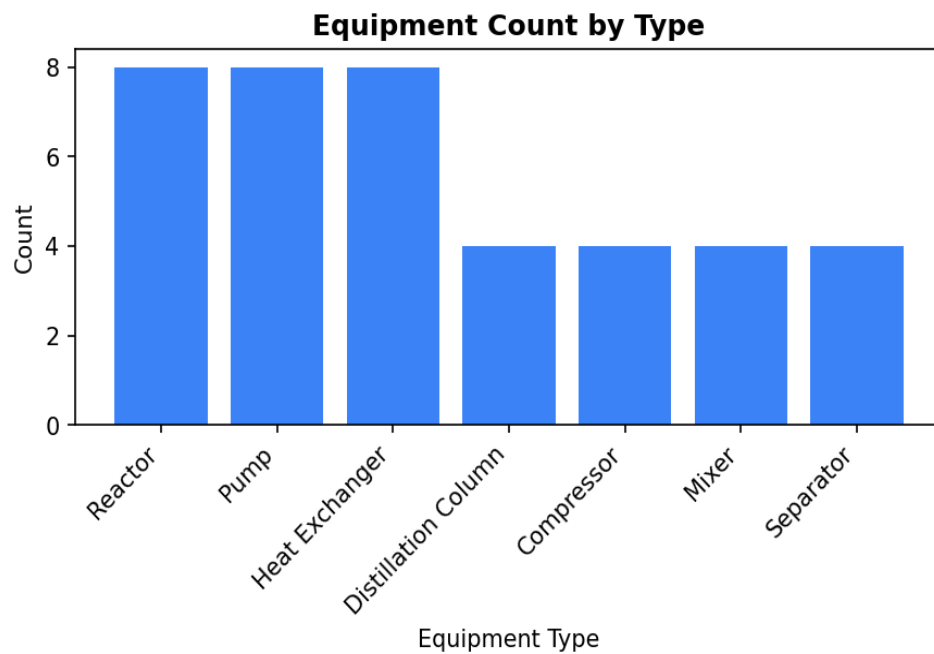
Equipment Type	Count	Percentage
Reactor	8	20.0%
Pump	8	20.0%
Heat Exchanger	8	20.0%
Distillation Column	4	10.0%
Compressor	4	10.0%
Mixer	4	10.0%
Separator	4	10.0%

Visual Analysis

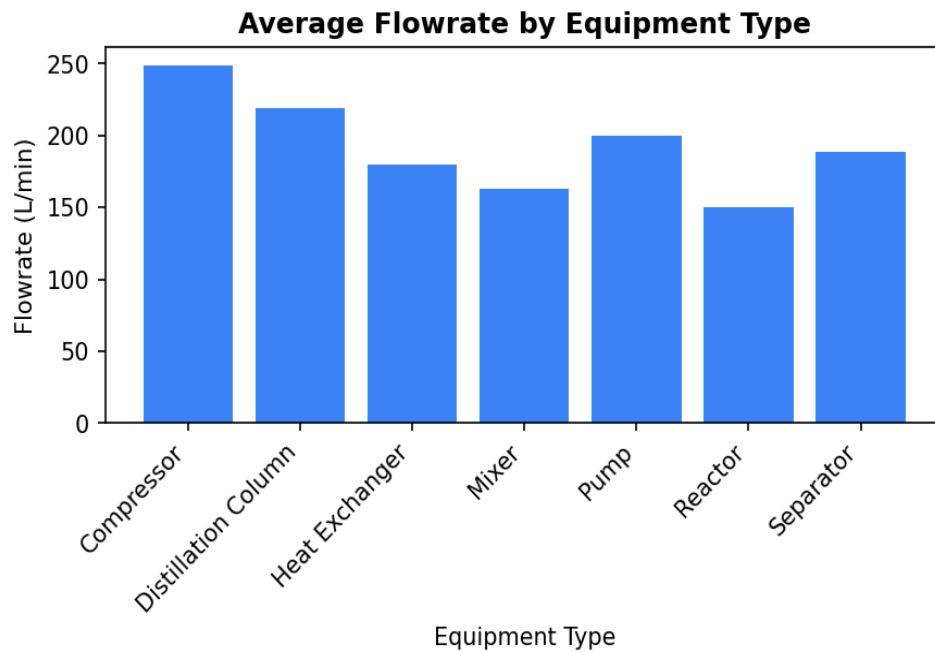
Equipment Type Distribution



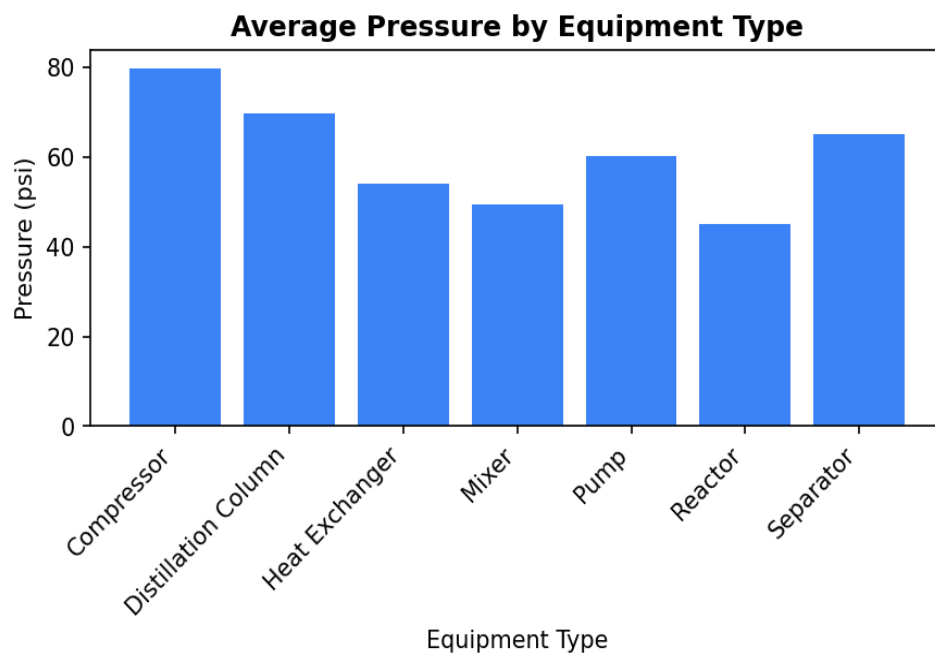
Equipment Count by Type



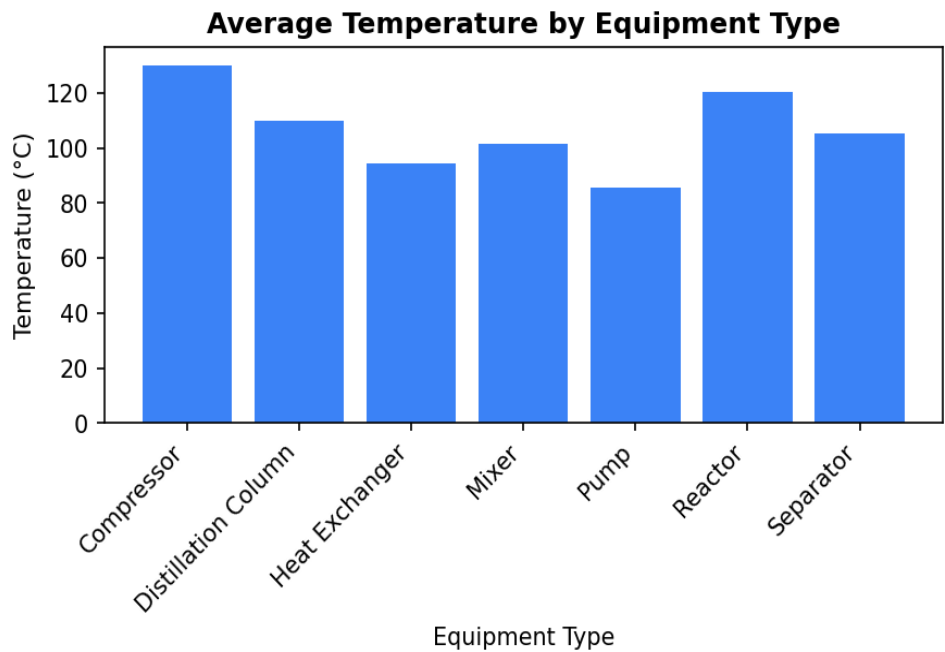
Average Flowrate by Equipment Type



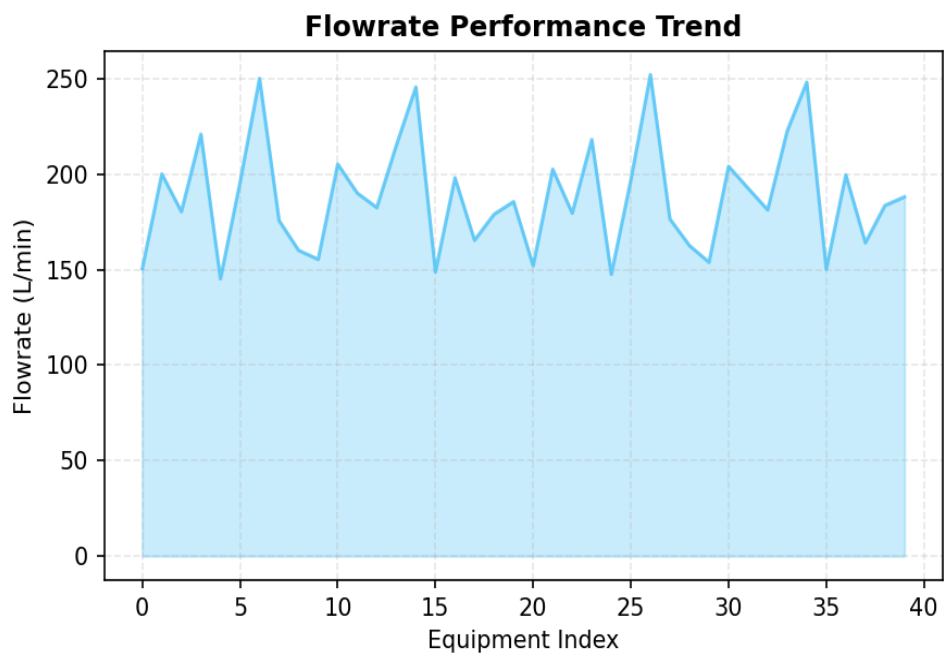
Average Pressure by Equipment Type



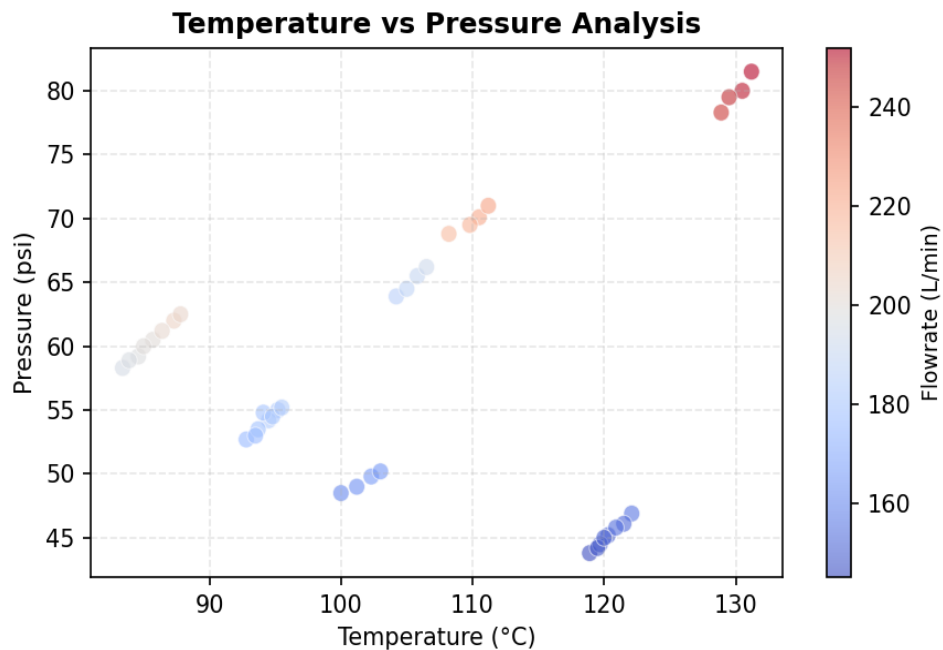
Average Temperature by Equipment Type



Flowrate Performance Trend (All Equipment)



Temperature vs Pressure Analysis (All Equipment)



Detailed Equipment Data

Equipment Name	Equipment Type	Flowrate (L/min)	Pressure (psi)	Temperature (°C)
Reactor-A1	Reactor	150.50	45.20	120.30
Pump-B2	Pump	200.00	60.50	85.70
Heat Exchanger-C3	Heat Exchanger	180.30	55.00	95.20
Distillation Column-D4	Distillation Column	220.80	70.10	110.50
Reactor-A2	Reactor	145.20	43.80	118.90
Pump-B3	Pump	195.50	58.30	83.40
Compressor-E5	Compressor	250.00	80.00	130.50
Heat Exchanger-C4	Heat Exchanger	175.60	52.70	92.80
Mixer-F6	Mixer	160.00	48.50	100.00
Reactor-A3	Reactor	155.30	46.90	122.10
Pump-B4	Pump	205.20	62.00	87.30
Separator-G7	Separator	190.00	65.50	105.80
Heat Exchanger-C5	Heat Exchanger	182.40	54.20	94.50
Distillation Column-D5	Distillation Column	215.00	68.80	108.20
Compressor-E6	Compressor	245.50	78.30	128.90
Reactor-A4	Reactor	148.80	44.50	119.70
Pump-B5	Pump	198.00	59.20	84.60

Equipment Name	Equipment Type	Flowrate (L/min)	Pressure (psi)	Temperature (°C)
Mixer-F7	Mixer	165.30	49.80	102.30
Heat Exchanger-C6	Heat Exchanger	178.90	53.50	93.70
Separator-G8	Separator	185.50	63.90	104.20
Reactor-A5	Reactor	152.00	46.10	121.50
Pump-B6	Pump	202.50	61.20	86.40
Heat Exchanger-C7	Heat Exchanger	179.50	54.80	94.10
Distillation Column-D6	Distillation Column	218.00	69.50	109.80
Reactor-A6	Reactor	147.50	44.20	119.50
Pump-B7	Pump	196.80	58.90	83.90
Compressor-E7	Compressor	252.00	81.50	131.20
Heat Exchanger-C8	Heat Exchanger	176.40	53.00	93.50
Mixer-F8	Mixer	162.50	49.00	101.20
Reactor-A7	Reactor	153.80	45.80	120.90
Pump-B8	Pump	204.00	62.50	87.80
Separator-G9	Separator	192.50	66.20	106.50
Heat Exchanger-C9	Heat Exchanger	181.20	54.50	94.80
Distillation Column-D7	Distillation Column	222.50	71.00	111.20
Compressor-E8	Compressor	248.00	79.50	129.50
Reactor-A8	Reactor	150.20	45.00	120.00
Pump-B9	Pump	199.50	60.00	85.00
Mixer-F9	Mixer	164.00	50.20	103.00
Heat Exchanger-C10	Heat Exchanger	183.50	55.20	95.50
Separator-G10	Separator	188.00	64.50	105.00

Total rows displayed: 40